

Abstract Book

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Alphabetical by author

Sampling-Based Regularisation for Adversarial Linear Bandits

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We study the design of sampling-based regularisation in sequential decision-making. In the online learning setting, we show that the exploration-exploitation trade-off can be optimally controlled by sampling from carefully designed probability distributions, providing a statistical alternative to explicit penalty functions. Focusing on adversarial linear bandits, we develop a unified framework that connects Follow-the-Regularized-Leader (FTRL) and Follow-the-Perturbed-Leader (FTPL), extending their known equivalence beyond the full-information setting. Within this framework, we introduce self-concordant perturbations, a family of distributions that mirror the role of self-concordant barriers used in the FTRL-based SCRiBLE algorithm. This yields a new FTPL algorithm that combines self-concordant regularisation with efficient stochastic exploration. The resulting method achieves regret $O(dn \ln n)$ on both the d -dimensional hypercube and the Euclidean ball. On the Euclidean ball, this matches the best known guarantees for self-concordant FTRL methods, while on the hypercube it improves them by a factor of d , matching the optimal rate up to logarithmic factors.

Parameter estimation for graphon-interacting particle systems from discrete observations

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In this paper, we address the problem of joint parameter estimation for the drift and diffusion coefficients of heterogeneously interacting diffusive particle systems and their associated large population limits. Unlike the homogeneous setting, the interaction here is of mean-field type with weights characterized by an underlying graphon. This heterogeneity introduces significant analytical complexity: while homogeneous particles converge to an i.i.d. limit, our system leads to particles that are independent but not identically distributed in the limit. Starting from discrete observations of the system over a fixed interval $[0, T]$, we propose a contrast function based on a pseudo-likelihood approach. We establish that the resulting estimators are consistent as the discretization step tends to zero and the number of particles tends to infinity. Furthermore, we prove asymptotic normality under an additional regime linking discretization and particle count, demonstrating that the graphon-based structure can be successfully navigated despite the lack of identical distribution in the limit.

An ordering for the strength of functional dependence

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We introduce a new dependence order—the conditional convex order—whose minimal and maximal elements characterize independence and perfect dependence. Moreover, it characterizes conditional independence, satisfies information monotonicity, and exhibits several invariance properties. Consequently, it is an ordering for the strength of functional dependence of a random variable on a random vector. As we show, various recently studied dependence measures—including Chatterjee’s rank correlation, Wasserstein correlations, and rearranged dependence measures—are increasing in this order and inherit their fundamental properties from it. We characterize the conditional convex order by the Schur order and by the concordance order, and we verify it in settings such as additive error models, the multivariate normal distribution, and various copula-based models. Our results offer a unified perspective on the behavior of dependence measures across statistical models.

Statistical foundations of representation learning in generative models

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One of the key paradigm shifts in statistical machine learning over the past decade has been the transition from handcrafted features to automated, data-driven representation learning. A crucial step in this pipeline is to identify latent and learn structured

representations from data. In many applications, meaningful concepts are not directly observed, and must be learned from data, often using flexible, nonparametric models such as deep generative models. These settings present new statistical and computational challenges that will be focus of this talk. We will re-visit the statistical foundations of nonparametric latent variable models as a lens into the problem of identifying representations in deep generative models. We discuss our recent work on developing methods for identifying and learning causal representations from data with rigorous guarantees, and discuss how even basic statistical properties are surprisingly subtle. Along the way, we will explore the connections between deep generative models, nonparametric latent variable models, and causal graphical models.

Reoptimization in the dynamic and stochastic knapsack problem: from near-optimality to distributional equivalence

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We study a dynamic and stochastic knapsack problem in which a decision maker is sequentially presented with n items with unitary rewards and independent weights that are drawn from a known continuous distribution. The decision maker seeks to maximize the expected reward she collects by including items in a knapsack while satisfying a capacity constraint, and while making terminal decisions as soon as each item weight is revealed. We propose a reoptimized heuristic and compare its total rewards with that of the optimal dynamic programming policy. We show that the two total rewards have the same asymptotic mean, the same asymptotic variance, and the same limiting distribution. In contrast, we also note that other asymptotically optimal heuristics that have been considered in the literature have different (larger) higher moments and different limiting distributions. (Joint work with Xinchang Xie and Yun-Tung Kuo.)

Percolation on Hierarchical Lattices

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In this talk we examine families of hierarchical lattices, with focus on its fine percolative properties. Thanks to the self-similarity of the graphs in these sequences, several properties of the model can be examined, including: critical exponents, scaling relations, and noise sensitivity. Furthermore, for these models, we obtain explicit expressions for several critical exponents of percolation. The results are very sharp and proofs employ a

combination of tools from probability and dynamical systems. Based on a joint work with Alves, Moreira, and Teixeira.

Numerical Radius of Non-Hermitian Random Matrices

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For a square matrix, the range of its Rayleigh quotients is known as the numerical range, which is a compact and convex set by the Toeplitz–Hausdorff theorem. The largest value in this convex set is known as the numerical radius, which is often used to study the convergence rate of iterative methods for solving linear systems. In this talk, we will introduce a recent result on the asymptotic behavior of the numerical radius of a large-dimensional, complex, non-Hermitian random matrix and its elliptic variants. For the former, remarkably, the radius can be represented as the extremum of a stationary Airy-like process, which undergoes a correlation-decorrelation transition from a small to a large time scale. Based on this transition, we obtain the precise first and second order terms of the numerical radius. In the elliptic case, we show that the fluctuation of the numerical radius reduces to the maximum of two independent Tracy-Widom variables. Based on joint work with Giorgio Cipolloni.

On the role of Rademacher complexities in statistical learning

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We study the problem of learning with respect to the squared loss over a convex class of functions. It has long been believed that the sample complexity in this setting is governed by localized Rademacher complexities. We show that, assuming access to coarse information on the covariance structure of the model class, the sample complexity is instead controlled by a localized complexity associated with the limiting Gaussian process. In heavy-tailed regimes, this quantity can be significantly smaller than the Rademacher complexity.

NEIGHBORHOOD ADAPTIVE ESTIMATORS FOR CAUSAL INFERENCE UNDER NETWORK INTERFERENCE

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Estimating causal effects has become an integral part of most applied fields. In this work we consider the violation of the classical no-interference assumption with units connected by a network. For tractability, we consider a known network that describes how interference may spread. Unlike previous work, the radius and intensity of the interference experienced by a unit are unknown and can depend on different local subnetworks and the assigned treatments. We study estimators for the average direct treatment effect on the treated in such a setting under additive treatment effects. We establish rates of convergence and distributional results. The proposed estimators consider all possible radii for each local treatment assignment pattern. In contrast to previous work, we approximate the relevant network interference patterns that lead to good estimates of the interference. To handle feature engineering, a key innovation is the use of synthetic treatments to decouple the dependence. We provide simulations, an empirical illustration, and insights for the general study of interference.

Cutoff for independent random walks on the circle conditioned not to intersect

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In this talk, we will consider a class of Markov processes on the discrete circle which has been introduced by König, O'Connell and Roch. These processes describe movements of exchangeable interacting particles and are discrete analogues of the unitary Dyson Brownian motion: a random number of particles jump together either to the left or to the right, with trajectories conditioned to never intersect. We provide asymptotic mixing times for stochastic processes in this class as the number of particles goes to infinity, under a sub-Gaussian assumption on the random number of particles moving at each step. As a consequence, we prove that a cutoff phenomenon holds independently of the transition probabilities, subject only to the sub-Gaussian assumption and a minimal aperiodicity hypothesis.

Emergence of host dormancy in the presence of a persistent virus epidemic

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We study a stochastic individual-based model for a microbial population challenged by a persistent virus epidemic. We assume that the resident microbial host population and the virus population are in stable coexistence upon arrival of a single new mutant host individual that is capable of switching into a reversible state of dormancy upon contact with virions as a means of avoiding infection. At the same time, we assume that this new

dormancy trait comes with a cost, namely a reduced individual reproduction rate while in the active state. We prove that the mutants can nevertheless invade the resident population with strictly positive probability in the large population limit. Given the reduced reproductive rate, such an invasion would be impossible in the absence of the virus epidemic. We explicitly characterize the parameter regime where this emergence of a host dormancy trait is possible, determine the success probability of a single invader and the typical amount of time it takes the successful mutants to reach a macroscopic population size. We conclude with an investigation of the fate of the population after the successful emergence of a dormancy trait. Here, either both host types and the virus will reach coexistence, or the mutant hosts will drive the resident hosts to extinction. This is joint work with András Tóbiás, Budapest.

Surrogate-powered Causal Inference

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Missing information is intrinsic to causal inference with survival outcomes: experiments are constrained by budgets and timelines, and observational studies face drop-out. In addition to the unobserved counterfactual outcome, the factual event time is subject to right censoring. This degrades the effective sample size even when identification is secured by design or by assumption. We study whether an intermediate post-treatment event, such as recurrence or progression, can recover the power for the primary outcome estimand that is lost to censoring. With full data and absent semiparametric restrictions, surrogate outcomes cannot deliver a first-order efficiency gain. With missing outcomes, surrogates and baseline covariates become informative for marginal causal parameters that, absent missing data, do not depend on the surrogate. The size of the gain depends on the balance of the post-treatment stratification induced by the surrogate, the prognostic separation of the strata, and the temporal proximity of the stratification with the censoring.

Calibration and minimizing loss

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Are calibration and minimizing loss compatible goals? Is calibration desirable for minimizing loss? Does minimizing loss lead to calibration? We study these questions via tradeoff and recalibration arguments and show that, all other things equal, the answer to all three questions is yes for various calibration modes and proper losses. But, in general, miscalibrated forecasts can be superior to calibrated ones, even across all proper losses at

the same time. Furthermore, excluding auto-calibration, the answer can be loss-dependent. We introduce a new family of calibration modes that generalize marginal calibration in a unique way for each proper loss and arise naturally from minimizing loss. This family enjoys explicit tradeoffs that are analogous to the bias-variance tradeoff in regression when bias is identified with miscalibration.

BabyTransformer CODL learns latent domains

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Token-level contextual embeddings for text corpora are now readily available for both natural language and genetic sequences. We take a top-down perspective and introduce CODL, a continuous topic model that leverages these embeddings to uncover the latent domains represented in a corpus. The model is a very large p -mixture of high-dimensional Gaussian distributions, where the mixture weights themselves are mixtures of K Softmax probability vectors supported on the p Gaussian means. By first learning the many latent Gaussian means from the tokenized corpus, we construct a corpus-specific vocabulary. This vocabulary forms the support of a discrete K -Softmax mixture ensemble whose atoms encode the corpus' latent domains in the embedding space. These learned softmax atoms can then be mapped back to natural language via the corpus-specific vocabulary, enabling interpretable topic discovery. A central component of our procedure is a new method for learning the latent domains from data, prior to full model fitting. We show that this reduces to detecting whether a Softmax mixture is primarily concentrated on a single leading component. To achieve this, we introduce a detection criterion that directly estimates functionals of a Softmax mixing measure, bypassing the challenging task of estimating the mixing measure itself in high dimensions. We provide end-to-end theoretical guarantees for EM-based atom estimates, and illustrate their interpretability by data examples.

Linear spreading speed of a branching annihilating random walk

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We study a branching annihilating random walk in which particles evolve on the lattice in discrete generations. Each particle produces a Poissonian number of offspring which independently move to a uniformly chosen site within a fixed distance from their parent's position. Whenever a site is occupied by at least two particles, all the particles at that site are annihilated. This can be thought of as a very strong form of local competition and implies that the system is not monotone. For certain ranges of the parameters of the

model, we show that the system dies out almost surely or, on the other hand, survives with positive probability. In an even more restricted parameter range, we strengthen the survival results to complete convergence with a non-trivial invariant measure, and in the same regime we show existence of a linear spreading speed for the model on the integer line. A central tool in the proofs is comparison with oriented percolation on a coarse-grained space-time lattice, using carefully tuned density profiles which expand in time and are reminiscent of discrete travelling wave solutions. Our results extend to possibly non-monotone models for which comparison with oriented percolation and other coupling techniques are possible. Based on joint works with Matthias Birkner, Jiří Černý, Nina Gantert and Pascal Oswald.

Casewise and Cellwise Robust Multilinear Principal Component Analysis

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Multilinear Principal Component Analysis (MPCA) is an important tool for analyzing tensor data. It performs dimension reduction similar to PCA for multivariate data. However, standard MPCA is sensitive to outliers. It is highly influenced by observations deviating from the bulk of the data, called casewise outliers, as well as by individual outlying cells in the tensors, so-called cellwise outliers. This latter type of outlier is highly likely to occur in tensor data, as tensors typically consist of many cells. This talk introduces a novel robust MPCA method that can handle both types of outliers simultaneously, and can cope with missing values as well. This method uses a single loss function to reduce the influence of both casewise and cellwise outliers. The solution that minimizes this loss function is computed using an iteratively reweighted least squares algorithm with a robust initialization. Graphical diagnostic tools are also proposed to identify the different types of outliers that have been found by the new robust MPCA method. The performance of the method and associated graphical displays is assessed through simulations and illustrated on two real datasets.

Doubly self-normalized change point test under local stationarity via sparse subsampling and stationary pooling

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This project studies change point tests for locally stationary time series based on sparse subsampling and stationary pooling. First, studentized data are sparsely subsampled in such a way that all selected observations are lagged by a divergent amount. Consequently,

a studentized sparse cumulative sum process is constructed for testing change points. The resulting test is proven to be asymptotically pivotal. This approach automatically transforms a change point test designed for stationary conditions into one suitable for local stationarity, while bypassing change point-sensitive bootstrapping or computationally intensive simulations. Second, the p-values computed from different studentized sparse cumulative sum processes initiated at various time points are shown to be approximately stationary and satisfy a functional central limit theorem. A method called stationary pooling is proposed to efficiently aggregate these approximately stationary p-values to enhance statistical power. Stationary pooling is shown to retrieve the change point detection power as in the stationary setting. Building on sparse subsampling and stationary pooling, we propose a doubly self-normalized test that employs self-normalization twice for two layers of aggregation: across sparsifying lags and along consecutive initial time points. Simulations demonstrate improvements in power, size accuracy, and computational efficiency.

Convergence of the EM algorithm beyond classical Gaussian regimes

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The EM algorithm has been widely studied theoretically for finite Gaussian mixture models. However, the practical use cases of the EM algorithm span beyond these specialized theoretical models. In this talk we will cover recent advances in theory of the EM algorithm beyond the classical setting in two primary regimes. First, we will show the advantages of generalizing the EM algorithm to the Federated Learning regime wherein data heterogeneity poses a significant challenge to the applicability of classical statistical methods. We characterize the convergence rate of the EM algorithm under all regimes of data heterogeneity under finite mixtures of linear regressions. We show that rather than being a bottleneck, in some cases, data heterogeneity can surprisingly accelerate the convergence of iterative federated algorithms. Secondly, we delve briefly into the behaviour of the EM algorithm under mixtures of non-Gaussian densities. We show in particular that even with slower tail decay, the convergence rates of the EM algorithm for mixture densities or linear regressions do not suffer significantly, providing the foundations for extending many existing results to a larger class of mixture models.

DiPMInd: Distance Profile based Mutual Independence testing for random objects

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We develop a novel unified framework for testing mutual independence among random objects in possibly different metric spaces. We introduce new measures of mutual independence and propose tests—including existing methods as special cases—that achieve minimax rate optimality and strong empirical power. The foundation is the new concept of joint distance profiles, which uniquely characterize the joint law of random objects under mild conditions. Our test statistics quantify the difference of the joint distance profiles of each data point with respect to the joint law and the product of marginal laws of the vector of random objects. To enhance power, we integrate this difference with respect to different measures and incorporate flexible data-adaptive weight profiles. We derive the limiting distribution of the test statistics under the null hypothesis of mutual independence and show that the proposed tests with certain weight profiles are asymptotically distribution-free if the marginal distance profiles are continuous. We also establish the consistency of the tests under sequences of alternative hypotheses converging to the null. For practical implementations, we use a permutation scheme to approximate p-values and prove that the permutation-based tests maintain type I error control under the null and achieve consistency under alternatives. We demonstrate the power of the proposed tests across various types of data objects through simulations and real data applications.

MOSAIC: a unified framework for lossless, one-shot, federated learning algorithms

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Multi-site studies are now central to biomedical research, but inference across sites remains challenging due to regulatory limits on sharing individual-level data, heterogeneous data distributions, sparse events in small centers, and the logistical burden of multi-round communication. In this talk, I introduce MOSAiC (Multi-site One-Shot Aggregation of Compressed Risk Functions), a unified framework for modern distributed research networks. Notably, MOSAiC reframes distributed learning as a mathematical problem of compressing and aggregating risk functions, leveraging recent advances in tensor networks—state-of-the-art tools for high-dimensional function approximation in scientific computing. Our MOSAiC enjoys four desirable properties that have never been achieved by any of the existing federated algorithms (except linear regressions): one-shot communication, lossless recovery of pooled-data estimates, inclusiveness of all sites irrespective of size or event rarity, and analytic submodel exploitability without re-querying partners. I will illustrate MOSAiC's validity and efficiency through applications in drug relabeling, drug repurposing, and post-market safety surveillance.

Recent progress on Talagrand's convolution conjecture

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In 1989, Talagrand conjectured that under the heat semigroup on the Boolean hypercube, any nonnegative function on the cube exhibits a uniform tail bound that is better than that by Markov's inequality. We discuss recent progress on the conjecture and the reverse heat process used in the proof.

Large field problem in coercive SPDE via scaling

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In this talk, I will show an approach to deriving a priori bounds for coercive SPDEs based on scaling. The basic idea is to first show bounds for the equation with a small noise and then rescale the bounds to a global scale. While many equations that the approach can handle have been treated recently with other methods, its advantages are that it is quite simple and allows one to state a single result that is applicable to a variety of equations, such as rough differential equations and parabolic/elliptic SPDEs. Based on joint work with Massimiliano Gubinelli.

The random XORSAT decimation process

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We analyse the performance of Belief Propagation Guided Decimation, a physics-inspired message passing algorithm, on the random k -XORSAT problem. Specifically, we derive an explicit threshold up to which the algorithm succeeds with a strictly positive probability that we compute explicitly, but beyond which the algorithm with high probability fails to find a satisfying assignment. In addition, we analyse a thought experiment called the decimation process for which we identify a non-reconstruction and a condensation phase transition. The main results confirm physics predictions that link the phase transitions of the decimation process with the performance of the algorithm, and improve over partial results from a recent article. Joint work with Arnab Chatterjee, Mihyun Kang, Lena Krieg, Maurice Rolvien, and Gregory B. Sorkin.

Large Deviations and Bahadur Efficiency for General Divergence-based Estimators

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Minimum divergence estimators have long been studied as robust alternatives to maximum likelihood methods. In this work we investigate, from a general perspective, the rare-event behavior of such divergence-based inferential methods. Specifically, we establish large deviation estimates for the probabilities that a divergence-based estimator deviates from the true parameter values, and we characterize the associated exponential rate of decay as the sample size tends to infinity. This decay is described via a large deviation rate function whose form depends on the choice of the divergence measure. We then apply these results to study Bahadur efficiency for general divergence estimators, thereby extending and complementing earlier work on the topic.

Assessing the Quality of Denoising Diffusion Models in Wasserstein Distance

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Generative modeling aims to produce new random examples from an unknown target distribution, given access to a finite collection of examples. Among the leading approaches, denoising diffusion probabilistic models construct such examples by mapping a Brownian motion via a diffusion process driven by an estimated score function. In this work, we first provide empirical evidence that denoising diffusion probabilistic models are robust to constant-variance noise in the score evaluations. We then establish finite-sample guarantees in Wasserstein-2 distance that exhibit two key features: they characterize and quantify robustness to noisy score estimates, and they achieve faster convergence rates than previously known results. Furthermore, the obtained rates match those known in the Gaussian case, implying their optimality.

A non-equilibrium equivalence of ensembles result via propagation of chaos

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We consider an alternative formulation of non-equilibrium stochastic dynamics referred to as Norton dynamics. A reference system is perturbed by an external forcing with varying intensity so that the value of a given observable is fixed. This amounts to constraining the dynamics to a level set of this observable. For observables that are linear functions of the empirical measure of the dynamics, this can be viewed as a constraint on the empirical mean of the dynamics. This leads us to consider mean-field interacting particle systems.

We show via propagation of chaos that an equivalence of ensembles holds between the Norton formulation of non-equilibrium dynamics and the standard formulation in which the intensity of the forcing is fixed.

Local e-values for intersection hypotheses

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The recent e-closure framework shows that multiple testing procedures controlling an expected loss can be understood through families of local e-values assigned to intersection hypotheses, yielding a unifying route to valid and often uniformly improved procedures. In this talk I will show new false discovery rate controlling tests for settings in which it was not known before that such a test would exist.

Rao-Blackwellized e-variables

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The Rao-Blackwell theorem is a cornerstone result in classical statistics. In this talk we present an analogous result for e-variables, namely that any e-variable can be Rao-Blackwellized into a new e-variable that, for every concave utility function, attains greater expected concave utility than the original. We also present a more sophisticated version for log-utility and illustrate the conceptual breadth of these Rao-Blackwell theorems with an example on testing the variance in a linear regression model.

Surviving from the tip of a cone

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In two-type first passage percolation on the square lattice, two entities compete to capture the sites of the lattice. The entities spread between nearest neighbor sites at times specified by random passage times associated with the edges. We consider the case when both types have the same passage time distribution, with one type starting at the origin and the other from an infinite cone with tip at the origin. We give a partial answer to whether the type starting at the origin can grow unboundedly, in terms of the slope of the cone.

Pairwise Comparison Metrics: AUC, Win Ratio, and the Problem of Non-Transitivity

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The area under the receiver operating characteristic curve is a standard measure of discrimination for risk prediction models. In this talk, we examine several challenges in the use of AUC, including its sensitivity to study design, the distribution of predictors, and the presence of competing events. We also investigate AUC and the Win Ratio as members of a pairwise comparison group of statistics and discuss the implications of non-transitivity for the interpretation of treatment effects, particularly in randomized trials and meta-analysis.

Efficient sampling with discrete diffusion models: Sharp and adaptive guarantees

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Diffusion models over discrete spaces have recently shown striking empirical success, yet their theoretical foundations remain incomplete. In this talk, we discuss the sampling efficiency of score-based discrete diffusion models under a continuous-time Markov chain formulation, with a focus on tau-leaping-based samplers. We establish sharp convergence guarantees for attaining small Kullback-Leibler divergence for both uniform and masking noising processes, and introduce an adaptive sampler whose convergence rate is governed by an intrinsic information-theoretic quantity.

Right-Sizing Communication and Recommendation Set Size in AI-Assisted Search

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We model the interaction between a user and an AI-driven recommendation system. The user initiates the process by conveying preference information through a costly and noisy message. The AI assistant, acting as a Bayesian agent, interprets the user's message to form a posterior belief about their preferences and make product recommendations. We study the high-dimensional setting and characterize how optimal message precision and recommendation set size depend on the cost parameters under two distinct distributions from which recommendations can be sampled.

Localization in Quantum and Classical Systems

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Random band matrices are Hermitian matrices with random entries supported in a band of width W around the diagonal. The eigenfunctions of such matrices are expected to decay exponentially at the scale W squared in dimension one, and exponentially in a constant times W squared in dimension two. In this talk I will discuss some recent progress on these problems and some open questions.

Parameter identification in linear non-Gaussian causal models under general confounding

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Linear non-Gaussian causal models postulate that each random variable is a linear function of parent variables and non-Gaussian exogenous error terms. We study identification of the linear coefficients when such models contain latent variables and allow for arbitrary non-linear latent confounding. Our main result is a graphical criterion that is necessary and sufficient for deciding the generic identifiability of direct causal effects, together with an algorithmic implementation of the criterion.

A Proof of The Changepoint Detection Threshold Conjecture in Preferential Attachment Models

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We investigate the problem of detecting and estimating a changepoint in the attachment function of a network evolving according to a preferential attachment model on n vertices, using only a single final snapshot of the network. We resolve the conjecture affirmatively, proving that detection is impossible if the change occurs at time n minus a little- o of the square root of n . Furthermore, we establish that estimating the changepoint with an error smaller than a little- o of the square root of n is also impossible.

Seeing through correlations: Disentangled feature importance

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Quantifying feature importance with valid statistical uncertainty is central to interpretable machine learning, yet classical model-agnostic methods often fail under feature correlation. We introduce Disentangled Feature Importance, a model-agnostic framework

that combines principled statistical inference with computational flexibility to provide robust and interpretable attributions under severe feature interdependence.

Inference-Time Alignment for Diffusion Models via Variationally Stable Doob's Matching

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Inference-time alignment for diffusion models aims to adapt a pre-trained reference diffusion model toward a target distribution without retraining the reference score network, thereby preserving the generative capacity of the reference model while enforcing desired properties at inference time. In this work, we introduce variationally stable Doob's matching, a framework for provable guidance estimation grounded in Doob's h-transform. We establish non-asymptotic convergence rates for the estimated guidance and for the generated distributions in Wasserstein distance.

Online policy learning and inference by matrix completion

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We take a collaborative-filtering approach for online decision-making based on collective preferences when personalized covariates are unavailable. Assuming low-dimensional latent features, we formulate the problem as a matrix completion bandit, propose a policy learning procedure, and develop an online debiasing method for inference with asymptotic normality guarantees.

Minimax Optimal Conditional Two Sample Test via Distance Profiles

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We study the problem of comparing the conditional distributions of two random variables given a confounding variable, where all variables take values in general metric spaces. We introduce a new discrepancy measure and a conditional two-sample test that avoids local smoothing, remains broadly applicable, and is minimax-rate optimal against a large class of alternatives.

Optimal binning for treatment effect heterogeneity

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I present a workflow for inference on treatment effect heterogeneity in clinical trials and observational studies. The approach begins with a nonparametric hypothesis test of homogeneity using a class of decision rules and proceeds by inverting the test to obtain the optimal decision rule with provable regret guarantees.

Generalised Bayesian model selection

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Strictly proper scoring rules are widely used in probabilistic forecasting and have recently become popular for inference without tractable likelihood functions. Here we explore a generalised Bayesian model selection framework for such models, proposing a path sampling estimator with sequential Monte Carlo and studying its consistency theoretically and empirically.

Yule's “nonsense correlation”: Moments, density, and tests of independence for non-stationary Gaussian processes

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This talk revisits Yule's classical question on the empirical correlation of two independent random walks, explaining recent results on the higher moments and limiting density of the statistic, as well as new statistical tests of independence for long-memory fractional Brownian motion paths.

High-dimensional learning dynamics of multi-pass Stochastic Gradient Descent in multi-index models

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We study the learning dynamics of multi-pass, mini-batch stochastic gradient descent for empirical risk minimization in high-dimensional multi-index models with isotropic random data. In a proportional asymptotic regime, we provide an asymptotically exact characterization of the coordinate-wise dynamics and compare them with stochastic modified equations and gradient flow.

Gaussian Free Field with Hard Wall Constraints From Discrete Trees to Planar Domains

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The Discrete Gaussian Free Field serves as a canonical model for random interfaces in statistical mechanics. When conditioned to stay non-negative on a domain—the hard wall constraint—the field undergoes entropic repulsion, lifting away from the wall to accommodate its fluctuations. I will present recent advancements that refine our understanding of this phenomenon on trees and in the two-dimensional lattice.

Learning the score under shape constraints

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Score estimation has recently emerged as a key statistical challenge due to its pivotal role in generative modelling via diffusion models. Motivated by these applications, we study the minimax risk of estimating the score function of a log-concave density and develop a locally adaptive, multiscale estimator that attains the corresponding upper bounds.

Selective Inference in Adaptive Trials

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Many clinical trials follow a design with multiple stages in which the recruitment of participants and administered treatments are chosen adaptively based on interim analyses. We advocate the use of selective inference methods to construct more powerful tests that control the selective Type I error and provide finite-sample valid inference for a wide range of trial designs without assumptions on the law of the outcomes and covariates.

Niche construction emerging from fast ecological and slow evolutionary and environmental timescales in individual-based models

Coralie Fritsch

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I will introduce an individual-based model of niche construction based on birth-death processes of interacting subspecies immersed in a slowly-varying environment influenced

by the population state. I will present a convergence result showing that the joint dynamics of species sizes on a logarithmic scale and the environment can be approximated by an explicit dynamical system in the large-population limit, and discuss applications to coexistence and immune escape.

**Eigenvector Ergodization in coupled non-Hermitian Quantum Chaotic Systems:
random matrix approach**

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We characterize quantitatively how the eigenvectors of two coupled non-Hermitian quantum chaotic systems, modeled by independent complex Ginibre matrices, spread over the full system as the interaction strength grows. The resulting asymptotic formula provides a full picture of gradual ergodization of eigenvectors across the transition regime.

Decentralised bipartite matching

Ayalvadi Ganesh

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Modern data centres connect tens of thousands of processors. We model the scheduling of communications between these processors as a bipartite matching problem. We present a class of probabilistic matching algorithms that are decentralised and only require each node to know its two-neighbourhood in the bipartite graph, and analyze their performance on a class of random graphs.

**Mean-Field Analysis of an Interacting Particle System on Dynamically Coevolving
Graphs**

Ankan Ganguly

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We present a simple interacting particle system with a dynamic interaction graph subject to bidirectional feedback: particle evolution depends on the underlying graph, and the graph dynamics depend on particle states. We introduce a sampling perspective that yields mean-field limits for the particles, graphon limits for the network, and additional limits that capture the joint distribution of the particles and graph.

Some result about opinion dynamics

Nina Gantert

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We explain results about opinion dynamics, more precisely about the averaging process on infinite graphs. Vertices have real-valued opinions and when they interact along edges, both vertices take the average value of the two endpoints of the edge. No prerequisites required. Based on joint work with Timo Vilkas.

Variance renormalisation of parabolic stochastic PDEs

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Scaling arguments give a natural guess at the regularity condition on the noise in a stochastic PDE for a local solution theory to be possible, but this is not always correct. We present recent progress on variance renormalisation and discuss strengths and weaknesses of approaches such as regularity structures, Itô calculus, and stochastic sewing.

Calibrated Multi-Level Quantile Forecasting

Isaac Gibbs
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I will discuss an online method for calibrating quantile forecasts at multiple levels simultaneously. The proposed lightweight procedure can wrap around any baseline sequence of point or quantile predictions to produce adjusted quantile forecasts that are guaranteed to be calibrated, even against adversarial distribution shifts, while keeping quantiles ordered and retaining strong no-regret guarantees.

Copula-based estimation of functionals of aggregated risks

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The interest in this talk is in aggregated random variables, where the dependency between the components is modelled via copulas. We are in particular interested in a general class of functionals of such an aggregated random variable, and estimation of these functionals. The separation of dependence structure and margins allows for highly adaptable estimation procedures, accommodating parametric and nonparametric methods for both components. The generality of the framework enables its application to diverse contexts, and its practical use is demonstrated in real-data applications.

Assessing Monotone Dependence: Area Under the Curve Meets Rank Correlation

Tilmann Gneiting

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The assessment of monotone dependence between two random variables is a classical problem in statistics and many application domains. This talk develops common population-level theory, estimators, and tests that unify and extend disconnected strands of literature, bridging continuous and dichotomous settings and applying to all linearly ordered outcomes.

Hierarchies of Calibration: Classification and Regression

Tilmann Gneiting

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This talk reviews and extends notions of calibration that have been proposed for classification and regression tasks, with emphasis on hierarchical relations between the various notions as they apply to real-valued, continuous, nominal, and binary outcomes. It also revisits associated decompositions of proper scoring rules and consistent scoring functions into measures of miscalibration, discrimination, and uncertainty.

Principles and Flexibility in Multiple Testing

Jelle Goeman

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Closed testing is known to underlie many multiple testing methods. We extend closed testing to a novel necessary and sufficient principle for all multiple testing methods controlling any expected loss. This e-Closure principle asserts that every multiple testing method is a special case of a generalized closed testing procedure based on e-values, and it applies in particular to the false discovery rate.

Simulation-Based Methods for Valid Inference from Privacy-Protected Data

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Data subject to disclosure control is becoming increasingly ubiquitous in the social sciences. This talk discusses a set of approaches that leverage simulation-based methods to enable valid statistical inference in settings where privacy-protection mechanisms

obscure the underlying confidential data, including data swapping and modern differential privacy mechanisms.

Causal Spatial Quantile Regression

Yan Gong

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Treatment effects in economic, environmental, and epidemiological applications often vary across space. We propose a semiparametric neural network-based causal framework leveraging deep spatial quantile regression to estimate spatial quantile treatment effects, together with an efficient adjustment procedure to mitigate the impact of spatial hidden confounders.

The distribution of partial sums of the Steinhaus function

Ofir Gorodetsky

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The Steinhaus function is a random, completely multiplicative function on the positive integers, whose values on primes are independent and uniformly distributed on the complex unit circle. We describe recent joint work in which the limiting distribution of the partial sums of the Steinhaus function was determined. The limiting distribution is Gaussian with random variance, where the variance is given by the total mass of a random measure arising as an instance of critical multiplicative chaos.

E-Values obtained by Conditioning on a Sufficient Statistic are Essentially Optimal

Peter Grünwald

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Whenever a composite null hypothesis admits a sufficient statistic, the conditional likelihood ratio provides an e-variable. We give a precise analysis of such conditional e-variables for exponential family nulls and alternatives and show that, under suitable conditions, they are asymptotically indistinguishable from the growth-optimal e-variable.

Distributionally robust risk evaluation with an isotonic constraint

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We propose a shape-constrained approach to distributionally robust learning that assumes the unknown density ratio between the target distribution and its estimate is isotonic with respect to a partial order. At the population level, we provide a solution to the resulting optimization problem, and at the sample level we establish consistency results for an empirical estimator of the target distribution.

Mean field particles system : propagation of chaos and concentration

Arnaud Guillin

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We review recent progress on sharp propagation of chaos for mean field particle systems, in particular new results stated in terms of Fisher information. We also prove concentration inequalities via autorefinement of Poincaré inequalities, with applications to mean field particle systems.

Generative Fiducial Models

Jan Hannig

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While generalized fiducial inference and its variants have yielded many theoretical and practical results for parametric inference and uncertainty quantification, applying them to generative models remains challenging. We identify key issues and propose a truncation-based method within generalized fiducial inference that provides valid coverage and improved generalization performance for generative models.

Planar percolation and the loop $O(n)$ model

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We show that, for any planar graph, a broad class of positively associated site percolation processes must contain zero or infinitely many infinite connected components. We then apply this percolation statement to the loop $O(n)$ model on the hexagonal lattice and obtain strong evidence for conformally invariant behavior in the scaling limit for a range of parameters.

Analysis of Multiple Outcomes in Contaminated Trials Reinforced With Validation Data

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This paper is concerned with estimation and testing for treatment effects with multivariate outcomes when imperfect diagnostic tools are used to classify subjects into different groups. We propose moment-based approaches for estimating and testing treatment effects, compare them with alternatives, and illustrate the methods with gene-expression data from the GALA II asthma study.

Critical behaviour for weight-dependent random connection models

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The random connection model is a versatile model encompassing various variants of continuum percolation. We introduce the model and present results for the critical percolation phase transition in high-dimensional Euclidean space and in hyperbolic space.

Universal prediction with extended ranks

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The accuracy of predictions derived from a regression model depends on how well the model mimics key aspects of the conditional distribution of an outcome given its features. We develop a single inferential framework that can represent these aspects for a wide range of conditional distributions, based on an unknown but arbitrary non-decreasing transformation of a linear regression model and a pseudo-likelihood using a partial ordering of the observed outcomes.

How does limma-trend work? An empirical partially Bayes perspective

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In high-throughput biology, limma-trend improves power by shrinking variance estimates toward an estimated mean-variance trend before computing p-values and applying false discovery rate control. We view limma-trend through the lens of empirical partially Bayes inference, derive formal statistical justification for a nonparametric generalization, and develop a new procedure that learns the full conditional distribution of variances rather than just a trend.

Local Fréchet Regression with Riemannian Manifold Predictors

Changjun Im

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We develop a local Fréchet regression framework for metric-space-valued responses with predictors on a Riemannian manifold. Our approach defines intrinsic local constant and local linear estimators through normal coordinates, logarithmic maps, and the Riemannian volume density, thereby respecting the geometry of the predictor space rather than relying on an extrinsic Euclidean representation.

Maximum information gain in kernel ridge regression

David Janz
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We establish matching lower and upper bounds for the maximum information gain in kernel ridge regression with covariates in the unit hypercube for families of kernels including the Matérn and squared exponential. Our results are based on elementary methods and avoid strong assumptions used in earlier work.

On the number of particles in Activated Random Walk and the Abelian sandpile

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We consider the number of particles in two different models of self-organised criticality: the Activated Random Walk and the Abelian sandpile model. For each setting, we derive asymptotic results for stationary particle counts and identify scaling laws connected to the underlying geometry and dynamics.

Dynamic Networks with Node Heterogeneity and Homophily

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We propose a framework for modeling node heterogeneity and link homophily in dynamic networks. The model accounts for homophily associated with observed and latent traits and introduces a normalized squared loss that enables stable parameter estimation in high-dimensional settings, together with rigorous theoretical guarantees.

Nonparametric hazard rate estimation with associated kernels: theory and application to a two-phase aging model

Sarah Kaakai

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I present a general theoretical framework for the nonparametric estimation of the intensity of random events using associated kernels whose shape adapts locally to the estimation point. I also apply these results to a two-phase aging model using experimental data on *Drosophila*, showing how associated kernels correct boundary bias and clarify biological mechanisms of aging.

BSNMANI: Bayesian scalar-on-network regression with manifold learning

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Brain connectivity analysis is crucial for understanding brain structure and neurological function. We develop BSNMANI, a Bayesian scalar-on-network regression model with manifold learning, that extracts shared connectivity structures and subject-specific network features, while jointly studying their association with clinical phenotypes and enabling outcome prediction.

Robust and Efficient Domain Adaptation via ϕ -Divergences

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We propose a principled and robust domain adaptation framework based on distributionally robust optimization using ϕ -divergences. The framework bypasses restrictive assumptions, naturally handles outliers and misspecification in both source and target domains, and comes with strong theoretical generalization guarantees.

Temporal connectivity of random geometric graphs

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A temporal random geometric graph is a random geometric graph in which all edges are endowed with a uniformly random time-stamp, representing the time of interaction between vertices. We determine a threshold for the existence of monotone increasing paths between all pairs of vertices and show that temporal connectivity appears at a significantly larger edge density than simple connectivity of the underlying random geometric graph.

Avoiding the Price of Adaptivity: Inference in Linear Contextual Bandits via Stability

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We discuss a regularized EXP4 algorithm for linear contextual bandits that provably satisfies the Lai–Wei stability condition, thereby enabling valid Wald-type inference with quantitative central limit theorem rates. The same algorithm also achieves minimax-optimal regret up to logarithmic factors, showing that stability and statistical efficiency can coexist.

Extended Fiducial Inference for Individual Treatment Effects via Deep Neural Networks

Sehwan Kim

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We develop extended fiducial inference, a scalable framework that revisits Fisher’s original goal of quantifying parameter uncertainty directly from observations using modern statistical computing. As an application, we propose a Double Neural Network method for individual treatment effect estimation that enables uncertainty quantification for deep neural networks under growing model complexity.

Long-Range Order in the Monomer Double-Dimer Model with Long-Range Interactions

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We study dimer and double-dimer models on $\mathbb{Z}^d$ -like graphs that allow long-range edges whose weights decay with distance. For a large class of such interactions, we prove that the monomer double-dimer model exhibits long-range order, implying uniformly positive monomer correlations in the dimer model and macroscopic loops in the double-dimer model.

Random graphs as models of quantum disorder

Antti Knowles

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A disordered quantum system can be modeled by a large Hermitian random matrix, and one of its most remarkable expected features is a localization-delocalization transition for the eigenvectors. I report on recent progress in analyzing the phase diagram for the Erdős–

Rényi random graph model, including the emergence of localized and delocalized phases separated by a mobility edge.

An algorithmic procedure for solving the generalized minimum information checkerboard copula problem

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We propose a generalization of the minimum information copula principle that allows the inclusion of additional constraints fixing certain higher-order margins of the copula. We prove uniqueness of the associated optimization problem under a natural condition and derive an IPFP-like procedure for solving its checkerboard approximation, together with convergence guarantees and numerical illustrations.

Privacy-Utility Tradeoffs of Noisy Stochastic Gradient Descent in High Dimensions

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We analyze noisy stochastic gradient descent through a diffusion perspective that captures both statistical risk evolution and privacy loss dynamics in high dimensions. Focusing on least squares with l_2 regularization, we provide a precise characterization of utility and privacy tradeoffs, including a variant that avoids explicit knowledge of gradient sensitivity.

Recent progress on graph bootstrap percolation

Brett Kolesnik

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In H -percolation, edges in a random graph are initially active, and additional edges become active if they complete a copy of a fixed graph H . In this talk, we identify the critical threshold for clique percolation and discuss recent progress for general H , resolving open problems in the area.

The eigenvalues of iid matrices are hyperuniform

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We consider the point process of eigenvalues of real or complex non-Hermitian matrices with independent, identically distributed entries. I will present a recent result showing that

this point process is hyperuniform, meaning that the variance in the number of eigenvalues in a subdomain is much smaller than the volume of the subdomain.

From Theory to Practice: inference beyond independence and CLT

Arun Kuchibhotla

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Construction of confidence intervals and hypothesis tests have long been based on central limit theorems for estimators and variance estimation, but such methods may fail to retain validity uniformly and under dependence. I will present a framework that builds on HulC and substantially expands its scope, including results under independence, dependence, and diverging dimensions, with examples for empirical processes and high-dimensional means.

Universality of first-order methods on random and deterministic matrices

Dmitriy Kunisky

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I will present recent results on the universal behavior of general first-order methods, a flexible class of iterative algorithms that generalizes approximate message passing. Using a moment method, we study these dynamics through traffic distributions of matrices and show how the approach unifies known limit theorems and extends them to matrices with less randomness, more complicated variance profiles, and some deterministic settings.

Minimum Norm Interpolation Under Gaussian Covariates

Gil Kur

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We introduce a new perspective on minimum-norm interpolators under Gaussian covariates by leveraging tools from high-dimensional geometry. We obtain localized bounds on shrinkage of the ground truth under isotropic Gaussian covariates and prove sharp rates for the mean squared error of the ℓ_1 minimum-norm interpolator using a purely geometric argument.

M-Estimation for Non-stationary Spatial Data

Daisuke Kurisu

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We investigate local linear quantile estimation for non-stationary spatial random fields and derive a spatial Bahadur representation for the proposed estimator. This provides a unified asymptotic framework and facilitates further inference, including simultaneous confidence bands and spatial hypothesis testing.

A Central Limit Theorem for sums of spatial block variables and some applications

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Many statistical inference methods for spatial data are based on sums of functions of blocks of spatial variables rather than individual observations. This work proves a general central limit theorem for such sums of block variables using a version of Stein's method, allowing both pure- and mixed-increasing domain spatial asymptotic structures, with applications to several important spatial inference problems.

A nonparametric test for survival curves combining the Mann-Whitney effect and an overlap index

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In survival analysis several methods are available for testing the identity of two survival functions. We provide a method for comparing two survival functions in a right-censoring framework that is particularly useful for crossing survival curves, while still retaining good power in a proportional hazards setting, by combining the Mann-Whitney effect with an overlap index.

Trustworthy scientific inference with neural density estimators in intractable likelihood settings

Ann Lee

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Scientific inference often involves parameters governing complex physical phenomena where the likelihood is implicit or intractable. I present new methodology that leverages neural density estimators to produce confidence regions with nominal frequentist coverage for any value of the unknown parameter, even with just one observation, while also yielding higher constraining power when the prior is informative.

Sato--Tate random multiplicative function

Sun Kai Leung

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Random multiplicative functions have become a central theme in modern probabilistic number theory. In this talk, I introduce the Sato--Tate random multiplicative function and outline its connections to arithmetic quantum chaos, with emphasis on the statistical behaviour of Hecke eigenforms.

Testing for correlation between network structure and high-dimensional node covariates

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In many application domains, networks are observed with node-level features, and a common problem is to assess whether nodal covariates are correlated with the network structure itself. We present four novel methods for addressing this problem, along with theoretical guarantees under low-rank latent space models endowed with high-dimensional node covariates.

Comparing groups of networks

Liza Levina
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Network-structured data arise in many applications, and hypotheses of interest often concern a subset of edges rather than the entire network. This talk introduces methods for mesoscale testing on networks and for estimating subgroup-specific structure while leveraging shared structure across all observed networks.

Graph Release with Assured Node Differential Privacy

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I will introduce GRAND, a network release mechanism that releases entire networks while ensuring node-level differential privacy and preserving structural properties. Under a broad class of latent space models, we show that the released network asymptotically follows the same distribution as the original network, and we evaluate the approach on both synthetic and real-world datasets.

Harnessing Synthetic Data from Generative AI for Statistical Inference

Xihong Lin

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We present a principled framework for incorporating synthetic data in downstream statistical analysis that ensures valid statistical inference even when generative AI models are misspecified. The proposed synthetic-data-assisted methods integrate observed and synthetic data, remain robust to misspecified black-box generative models, and can improve inferential power when the generative models are informative.

Global dynamics and weak universality for the fractional hyperbolic Φ_4 -3-model

Ruoyuan Liu

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I consider the three-dimensional stochastic damped fractional nonlinear wave equation with cubic nonlinearity, also known as the fractional hyperbolic Φ_4 -3-model. I will discuss global well-posedness, invariance of the corresponding Gibbs measure, and a weak universality result across the full range of relevant fractional orders.

Flexible Selective Inference with Flow-based Transport Maps

Sifan Liu

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We propose a new framework for selective inference that does not require an exact analytic characterization of the selection event. The core idea is to learn a transport map, parameterized by a normalizing flow, that pushes forward a simple reference distribution to the conditional distribution given selection, thereby enabling valid p-values, confidence intervals, and likelihood-based inference for complex adaptive procedures.

Preferential attachment trees with vertex death: The super-linear regime

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We investigate an evolving random tree model in which new alive vertices are added preferentially and existing alive vertices may be killed according to another preferential rule. In the super-linear regime, we analyze structural properties of the limiting infinite tree and discuss conditions under which distinct limiting behaviors arise.

Interacting point processes: How does the mean field limit help us for statistical inference on them?

Eva Löcherbach

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We consider mean field interacting Hawkes processes composed of many components, observed over a fixed time interval. I will discuss how passing to the mean field limit can help address statistical questions such as estimation of interaction structure, detection of synchrony in spiking behavior, and inference on the underlying jump rate function, with applications to neuroscience.

A conversion theorem and minimax optimality for continuum contextual bandits

Karim Lounici

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We study contextual continuum bandits where a learner observes a context vector and selects an action in a convex set to minimize a context-dependent function. We establish a general static-to-contextual regret conversion theorem, derive minimax-optimal contextual regret rates in several settings, and show that sublinear contextual regret may be impossible without continuity in the context.

Robust, sub-Gaussian mean estimators in metric spaces

Gabor Lugosi

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We study the performance of estimators of the Fréchet mean in general metric spaces under possibly heavy-tailed and contaminated data. We propose a general estimator based on high-dimensional extensions of trimmed means, prove performance bounds that generalize optimal Euclidean results, and apply the theory to spaces with curvature bounded from below and to uniformly convex Banach spaces.

Conformal Prediction for Dyadic Regression

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We consider uncertainty quantification for dyadic regression models using conformal prediction. We establish finite-sample validity under various sampling mechanisms based

on joint exchangeability, and show that under certain conditions it is possible to construct asymptotically valid prediction sets for a missing entry under structured missingness.

On the satisfaction frequency of spectral characterization conditions

Nikita Lvov

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We give the first specific conjectures on how frequently graphs satisfy sufficient conditions for being uniquely characterized by spectral information. These conjectures arise from a theoretical framework based on abstract-algebraic random matrix statistics and reformulations of known spectral conditions in terms of module structure associated with adjacency matrices.

HeteroJIVE: Weighted Spectral Estimation for Shared Subspace Recovery under Multi-View Heteroskedasticity

Zhongyuan Lyu

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We propose HeteroJIVE, a weighted two-stage spectral algorithm for recovering shared low-dimensional subspaces across multiple related data matrices exhibiting both statistical and structural heterogeneity. We establish oracle-optimal weighting schemes, explicit error bounds, and show how the method overcomes previously identified non-diminishing error barriers.

Edgeworth expansion on Wiener chaos

Paul Mansanarez

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We investigate Edgeworth expansions for functionals of Gaussian fields that are not necessarily sums of independent variables. Using the Malliavin–Stein method and semigroup analysis, we derive total variation bounds between the law of a functional and its Edgeworth expansion, with rates depending on the fourth cumulant.

The Functional Graphical Lasso

Tomas Masak

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We consider the problem of recovering conditional independence relationships between jointly distributed Hilbertian random elements in the sparse high-dimensional regime. In this context, we propose an infinite-dimensional generalization of the graphical lasso, prove model selection consistency under natural assumptions, and extend many classical results to infinite dimensions.

Trimming and constraints based robust clustering for matrix-variate data

Agustín Mayo-Iscar

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We design a robust clustering estimator for data coming from a mixture of matrix normal distributions, based on trimming and eigenvalue constraints. The resulting method generalizes robust covariance ideas to matrix-valued data and is shown empirically to perform well on artificial and real datasets.

What we can know from complex functional data

Ian McKeague

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This talk discusses a nonparametric inference framework for functional data having sample paths of bounded variation, with applications to complex settings such as wearable device data and possibly non-smooth metric spacetime. The framework highlights what can be learned from such data and explores its methodological and applied implications.

Universality for the global spectrum of random inner-product kernel matrices

Benjamin McKenna

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We study random inner-product kernel matrices, an important class of nonlinear random matrices motivated by machine learning. We show that the global spectrum of such matrices is universal, holding for much more general randomness and nonclassical aspect ratios than previously known.

A phase transition for the corank of random band matrices over $\mathbb{F}_p$

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For the corank of random matrices over the p -element field, there is a widespread universality phenomenon. We determine the critical band width at which this universality breaks down for uniform random band matrices, motivated by analogous phase transitions in classical random matrix theory.

Moments of Exponential Functionals of Lévy Processes and Bernstein-Gamma Functions

Martin Minchev

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Exponential functionals of Lévy processes are classical objects in probability with applications ranging from self-similar Markov processes to mathematical finance. I will discuss their moments and Mellin transforms, explain the role of Bernstein-Gamma functions, and highlight several open questions on moment determinacy and possible matrix-valued extensions.

Learning Multi-Index Models in High-Dimensions

Theodor Misiakiewicz

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We study the problem of learning multi-index models where the label depends on a high-dimensional input only through an unknown low-dimensional projection. Using harmonic-analytic tools and lower bounds in the statistical query and low-degree polynomial frameworks, we derive sharp statistical and computational complexity bounds and construct spectral algorithms that nearly achieve them.

Statistics for regularized optimal transport with decreasing regularization

Gilles Mordant

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We discuss the statistical fluctuations of entropy-regularized optimal transport between empirical measures when the regularization parameter decreases with sample size. The talk highlights how dual potentials relate to the underlying density of the data and how regularized optimal transport connects to diffusions.

Stochastic control and reinforcement learning for greedy dynamic matching models

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We consider a dynamic matching model on a general graph in which incoming items must be matched immediately whenever possible. Using dynamic programming and reinforcement learning, we study optimal control and access control for such greedy matching systems, including threshold-type policies and policy-gradient methods.

Geodesic Causal Inference

Hans-Georg Müller

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We introduce a general framework for causal inference when responses reside in geodesic metric spaces, based on a new geodesic calculus and the concept of geodesic average treatment effect. Using Fréchet regression, we obtain doubly robust estimation together with consistency and convergence rates, and discuss extensions to other causal designs.

ANOVA-type Wasserstein tests for factorial designs

Axel Munk

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We introduce a general optimal transport framework for testing statistical hypotheses on probability measures supported on finite spaces. Inspired by analysis of variance, the resulting tests handle factorial designs and are based on pairwise comparisons of optimal transport distances and corresponding barycenters, with asymptotic distributions, computational methods, and applications.

The Central Role of Proper Scoring Rules in Modern Statistics: Theory, Robustness, and Applications to Complex Data

Monica Musio

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Proper scoring rules provide a foundational framework for evaluating probabilistic forecasts and constructing tailored inferential procedures. This talk reviews their theoretical foundations and practical relevance, emphasizing connections with M-estimation, robustness, compositional data analysis, and extreme-value applications.

Random polynomials: old and new

Hoi Nguyen

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The study of zeros of random polynomials remains an active area of research. I will discuss recent developments on universality, global and local correlations of zeros, concentration, variance of the number of real zeros, and central limit theorem fluctuations.

Random polynomials with dependent coefficients

Oanh Nguyen

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Classical results on random polynomials largely focus on independent coefficients. In this talk, we explore models with dependent coefficients, including weak dependence and Ising-type interactions, highlighting both the challenges they pose and the techniques used to analyze them.

Functional AMUSE for multivariate functional separable time series

Klaus Nordhausen

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We extend second-order source separation to multivariate functional stationary time series. The corresponding model is introduced, its identifiability properties are discussed, and the classical AMUSE estimator is generalized to the functional setting under a separable covariance operator assumption, with performance investigated in simulations.

Functional Metric Learning for Classification Problems

Todd Ogden

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We propose a functional metric learning framework that integrates functional principal component analysis with large margin nearest neighbors to define and learn an optimal semimetric tailored for functional data. The method accounts for structure, continuity, and noise in one- and multidimensional functional data, and demonstrates improved classification performance in simulations and EEG applications.

Modelling multiplex networks

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This talk discusses how to model and make inference for multiplex networks, using multivariate graph limits and decorated graphons as flexible limiting objects. I will highlight

choices of parameterization and summary statistics, as well as benefits and challenges that arise when working with multiplex observations.

On the convex hull and coverage threshold of Ginibre eigenvalues

Mohammed Osman

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We study two geometric properties of Ginibre eigenvalues: the convex hull and the coverage threshold, the latter being the radius for which the union of disks centered at each eigenvalue covers a given region. We obtain limit theorems for the perimeter, area, number of vertices of the convex hull, and the coverage threshold of disks.

A weak transport approach to the Schrödinger-Bass problem

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I explain how the dynamic Schrödinger-Bass problem admits an equivalent static formulation as a weak optimal transport problem with an explicit cost. This reformulation yields existence and duality results, provides a structural characterization of the optimal semimartingales, and naturally leads to a Sinkhorn-type iteration for computation.

State evolution beyond first order methods

Ashwin Pananjady

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We study iterative optimization algorithms applied to high-dimensional random data beyond the first-order setting. We develop a rigorous state evolution for a much larger class of algorithms, show that it can be written in a canonical form, and provide a fully non-asymptotic analysis using a sequential Gaussian comparison approach.

Hierarchical Clustering with Confidence

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We show how randomizing hierarchical clustering can be used not only to assess stability, but also to enable valid hypothesis testing based on the clustering results. The resulting nodewise tests control the Type I error rate for any hierarchical linkage and can also be used to estimate the number of clusters with a guarantee against overestimation.

From Atmospheric Rivers to Flood Risk: A Multivariate Model for Extreme Precipitation

Anna Panorska

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We propose a multivariate model for extreme precipitation events that describes their duration, intensity, and maximum, and links them to atmospheric flow patterns such as Atmospheric Rivers. The ultimate goal is to estimate the probabilities of such extreme events using statistical and meteorological tools.

Application of control theory to online parameter estimation

Matthew Parry

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We develop a learning framework based on proportional–integral–derivative controllers for online parameter estimation. The integral component removes persistent error while the derivative component helps damp over-correction, yielding a stable and robust approach with connections to momentum and adaptive optimization methods.

Revisiting M-estimation: risk bounds, optimality results, and probabilistic connections

Reese Pathak

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We discuss recent results on least squares estimation and empirical risk minimization, including risk bounds, deviation inequalities, and optimality and suboptimality results that improve classical guarantees. We also highlight new connections between the analysis of these estimators and geometric quantities such as Gaussian width.

DNA replication and cell division cycles coordination in *E. coli*: a mathematical approach to cell physiology

Alexandre Perrin

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Despite extensive research, the quantitative principles that govern the coordination between DNA replication and cell division in bacteria remain debated. This talk presents a unifying mathematical framework that compares single-process and double-process regulatory models, identifies necessary and sufficient conditions for physiological

behavior, and introduces a new model that robustly captures replication-division coordination across growth regimes.

A class of continuous-time Gaussian mean-variance mixture state-space models simultaneously accounting for asymmetry, tailedness, and peakedness

Krzysztof Podgorski

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This talk introduces a class of infinitely divisible mixing distributions for Gaussian mean-variance mixtures that retains flexibility while being closed under convolution. The resulting continuous-time Gaussian-IGG models support state-space formulations and accommodate asymmetry, peakedness, and heavy tails, with applications to stochastic volatility.

SYNTAX-GUIDED DIFFUSION LANGUAGE MODELS WITH USER-INTEGRATED PERSONALIZATION

Annie Qu

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We develop a syntax-guided diffusion language model that incorporates latent syntactic variables to enhance text quality, diversity, and personalized conditioning. The proposed framework introduces syntax-to-text generation, shared latent representations across users, and interpretable control over both syntactic and lexical components, with strong empirical performance in fluency, diversity, and stylistic fidelity.

Community Detection and Modularity in the Preferential Attachment Block Model

Miklos Racz

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I will discuss community detection in the preferential attachment block model. The key to the results is a detailed understanding of modularity of partitions in the model, which may be of independent interest and underpins new results on community detection.

A Minimum Covariance Determinant Approach for Matrix and Tensor Data

Una Radojicic

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This work introduces the Matrix Minimum Covariance Determinant, a robust estimator for location and covariance tailored to matrix-valued data that preserves matrix structure and extends naturally to tensors. It provides matrix-affine equivariance, strong robustness properties, efficient algorithms, and interpretable decompositions based on Shapley values for outlier analysis.

Elements of Interactive Decision Making

Alexander Rakhlin

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Machine learning methods are increasingly deployed in interactive environments, ranging from dynamic treatment strategies in medicine to reinforcement-learning-based fine-tuning of large language models. We present a general framework for interactive decision making that subsumes bandits and reinforcement learning, with a focus on characterizing sample complexity and identifying core algorithmic primitives.

Quantitative Unimodular Propagation of Chaos and its Consequences

Kavita Ramanan

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We establish quantitative versions of unimodular propagation of chaos for interacting particle systems on sparse random graphs and describe some of the resulting consequences. In particular, the local-field equation emerging from this theory naturally interpolates to the classical mean-field equation as the graph size grows.

Assumption-lean weak limits and tests for two-stage adaptive experiments

Zhimei Ren

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Focusing on two-stage adaptive experimental designs, we derive new weak convergence results for weighted inverse probability weighted estimators under significantly weaker assumptions than previous work. We also propose a computationally efficient simulation-based method for inference when limiting distributions are non-normal.

Robust and Sub-Gaussian Estimation of Means and Covariances

Zoraida Rico

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We revisit trimmed mean estimators and show that they achieve optimal performance for one-dimensional mean estimation with strong central limit behavior. We then extend these ideas to covariance estimation in high dimensions under heavy tails and contamination, obtaining optimal dimension-free rates of convergence.

A stochastic continuous-space model and its large population limit for the dynamic of actin polymerisation and depolymerisation in the cell

Timothé Ringear

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We model actin monomers in cells by a stochastic process in continuous space where diffusing monomers can agglomerate into polymers at prescribed locations. In a large population limit, the normalized system converges to a deterministic partial differential equation with a singularity at the polymer location.

The distributionally robust prediction error of the square root LASSO and related estimators

Cynthia Rush

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We study linear predictors generated by the square root LASSO and related estimators and provide conditions under which they minimize worst-case prediction error over a Wasserstein-type uncertainty set. The analysis yields a practical recommendation for the regularization parameter and gives out-of-sample performance guarantees without relying on sparsity assumptions.

Chaos and Riemann zeta function on the critical line

Eero Saksman

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We discuss results, both old and new, on the statistics of the Riemann zeta function on short intervals on the critical line. The talk highlights links to random multiplicative chaos and recent collaborative progress on this classical object of analytic number theory.

Learn the score

Richard Samworth

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Score estimation has recently emerged as a key modern statistical challenge, due to its role in diffusion models and convex M-estimation for linear regression. We study minimax risk under log-concavity, define meaningful subclasses that capture tail behavior and smoothness, and construct locally adaptive estimators that attain the resulting upper bounds.

Provable false discovery rate control for deep feature selection

Kazuma Sawaya

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We develop a flexible feature selection framework based on deep neural networks that approximately controls the false discovery rate. The method applies broadly across architectures, including multilayer perceptrons, convolutional and recurrent networks, attention mechanisms, and residual connections, and comes with theoretical guarantees under a high-dimensional multi-index model.

Optimal Transport for Inverse Problems

Carola-Bibiane Schönlieb

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Optimal transport has emerged as a powerful framework for comparing structured data distributions and has recently shown promise in inverse problems. I will discuss how transport-based discrepancies, priors, and geometries can improve robustness, reconstruction quality, and interpretability in inverse imaging problems, especially under sparse, noisy, or distribution-shifted data.

The contact process on dynamical random trees with degree dependence

Marco Seiler

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We study the contact process on dynamically evolving Bienaymé-Galton-Watson trees where edges open and close at degree-dependent rates. We determine how the update speed and connection probabilities affect the existence of a non-trivial survival phase transition and provide conditions under which the phase transition is trivial.

Wasserstein-Cramer-Rao Theory of Unbiased Estimation

Bodhisattva Sen

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We develop a theory of sensitivity of unbiased estimators based on Wasserstein geometry, paralleling the classical Cramér-Rao theory based on Fisher-Rao geometry. This yields a Wasserstein-Cramér-Rao lower bound, exact characterizations in special models, and asymptotic optimality results for Wasserstein projection estimators across a range of statistical examples.

Spectral algorithms for signal recovery from multi-modal data

Subhabrata Sen

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We study spiked Wigner matrices with correlated spikes, motivated by multimodal principal component analysis. We derive and analyze a spectral algorithm for signal recovery based on a linearization of approximate message passing and identify a sharp signal-to-noise threshold beyond which an informative outlier eigenvalue emerges.

Generalization via distributional learning

Xinwei Shen

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This talk focuses on generalization beyond the observed data distribution, including extrapolation, distribution shifts, and causal inference. We propose tackling such problems through a distributional perspective via engression, a method that blends generative models with proper scoring rules to improve out-of-distribution generalization.

Calibration in multiple hypothesis testing

Jake Soloff

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This talk develops new perspectives on empirical Bayes and compound-decision inference for local false discovery rate criteria. It introduces a simple nonparametric procedure that exactly controls the expected maximum local false discovery rate among discoveries and connects multiple testing with probabilistic forecasting through a prior-free interpretation of local false discovery rates.

Doubly regularized generalized linear models for spatial observations with high-dimensional covariates

Arjun Sondhi

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We develop a doubly regularized regression framework for spatially correlated outcomes observed on a lattice with high-dimensional covariates and a second network structure on the features. The resulting estimators can be implemented with standard convex optimization algorithms and support asymptotically valid confidence intervals and hypothesis tests.

Local inference for functional time series

Helle Sørensen

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We consider local inference for functional time series, where hypotheses are tested along the domain of the functional observations in the presence of dependence. The proposed tests are based on permutations, address dependence in the time series and fixed effects, and are applied to daily ice-cover data in the Baffin Bay.

Spectral regularized kernel two-sample tests

Bharath Sriperumbudur

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We study the optimality of kernel-based two-sample tests on general domains. We show that the popular maximum mean discrepancy test is not optimal in Hellinger separation, propose a spectrally regularized modification that is minimax optimal, and develop an adaptive version that remains nearly minimax optimal while supporting permutation calibration.

Spectral methods for graph inference: the case of community detection

Ludovic Stephan

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Community detection is the task of recovering hidden cluster structure from network information. I will present several spectral methods that address the failure of adjacency-based algorithms in sparse graphs, including the celebrated non-backtracking method, its variants, and the Bethe Hessian.

Discussion on Open Questions Involving Moments and Cumulants

Jordan Stoyanov

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This discussion highlights open problems related to moment and cumulant determinacy, infinite divisibility, transformed distributions, and explicit constructions of continuous and discrete distributions sharing the same moments. It invites broader engagement with unresolved questions arising from recent research on moments, cumulants, and Stieltjes classes.

Optimality of Self-distillation in Spiked Covariance Models

Pragya Sur

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We present a rigorous framework for analyzing a broad class of spectral-shrinkage estimators in high dimensions and show that, in spiked covariance models, self-distillation achieves uniquely optimal prediction and estimation performance. We also discuss a federated learning setting where multiple centers share self-distilled estimators and optimal aggregation strategies can be characterized.

Model-oriented distances between causal graphs via posets

Armeen Taeb

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We propose a model-oriented framework for defining distances between graphs that represent statistical models. By organizing graph spaces as partially ordered sets based on model inclusion, we define a shortest-path metric that better reflects the structure of the model space than purely combinatorial alternatives.

Selection of the fittest or selection of the luckiest: the emergence of Goodhart's law in evolution

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Using a stochastic model in which selection acts on noisy phenotypic expressions rather than directly on genetic fitness, we show that increasing selection strength accelerates adaptation only up to a critical threshold. Beyond this point, stronger selection paradoxically slows evolution and erodes diversity by favoring the luckiest rather than the genetically fittest.

Consensus dimension reduction via multi-view learning

Tiffany Tang

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Different dimension reduction methods often produce conflicting visualizations of the same high-dimensional data. We propose a consensus approach that summarizes multiple visualizations into a single, more robust low-dimensional representation by leveraging ideas from multi-view learning to extract the shared structure across methods and hyperparameter choices.

Post-reduction inference for confidence sets of models

Yanbo Tang
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In high-dimensional regression problems, it is often more informative to construct a set of plausible explanatory models than to select a single model. We present a conditional-inference-based approach for constructing such confidence sets without sample splitting, illustrating the method in normal linear regression and extensions to time-to-event and more general regression models.

Return probabilities on Bienaymé-Galton-Watson trees

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We derive optimal upper bounds for annealed return probabilities of simple random walk on Bienaymé-Galton-Watson trees for all offspring distributions with finite first moment. The proof controls regions with poor isoperimetric properties and yields applications to the spectral theory of Erdős–Rényi random graphs.

Gradient equilibrium, Blackwell approachability, and ergodicity

Ryan Tibshirani
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Gradient equilibrium is a property of online learning iterates under which the average gradient converges to zero. I will review its basic theory and describe new connections to Blackwell approachability and ergodic theory, highlighting implications for sequential calibration and unbiasedness.

Selective and marginal selective inference for exceptional groups

Surya Tokdar

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We study confidence interval procedures for selected populations in multipopulation studies, showing that exact selective coverage control can lead to infinite expected width. We develop selection-adjusted empirical Bayes confidence procedures that approximate oracle selective intervals while maintaining finite expected width and favorable performance in numerical comparisons.

Smooth Zero-Inflated Modeling on Counting Tensors

Elena Tuzhilina
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We develop a tensor-based imputation framework for single-cell Hi-C data using a zero-inflated Poisson model to distinguish true and false zeros. The method enables efficient parameter estimation and improves recovery of missing chromatin contact information across cell types.

Change-point detection in preferential attachment models

Remco van der Hofstad
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We discuss the problem of identifying a change point in the attachment mechanism of affine preferential attachment models, where the constant in the attachment rule changes during network evolution. We give conditions under which such a change can be detected, with precision close to optimal.

One-arm exponents of the Ising model

Diederik van Engelenburg
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I will focus on the behavior of the Ising model in high dimensions. Recent work shows that the one-arm exponent and even the upper critical dimension depend on the boundary condition. In particular, the one-arm probability decays with different exponents under wired boundary conditions and in infinite volume, revealing a sharp contrast in critical behavior.

On an extension of the Cox model for time-varying covariates under dependent censoring with unknown association

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This paper proposes a copula-based Cox model to account for dependent censoring in the presence of time-varying covariates when the copula parameter is unknown. By estimating the copula parameter from the observed data and expanding the nonparametric baseline hazard via spline bases, the method enables maximum likelihood and penalized likelihood inference with stable variance estimation and reduced sensitivity to knot placement.

On the spectral determinacy of random graphs

Alexander Van Werde
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How much about a network can be deduced from the eigenvalues of associated matrices? I discuss old and new conjectures as well as recent progress on whether Erdős–Rényi random graphs are uniquely characterized by their adjacency spectra, with a focus on spectral determinacy in the typical case.

Real-time anomaly detection in space-time processes via conformal prediction and functional data analysis

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We develop an anomaly detection method for temporally evolving surfaces observed over two-dimensional domains, combining functional autoregressive modeling with conformal prediction. The method yields spatially and temporally varying prediction ranges with exact control of joint false alarm probabilities across the spatial domain and is applied to satellite measures of land elevation speed.

A "piecewise Gaussian" limit for the fluctuations of a stochastic spatial chemical reaction network

Amandine Véber
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We introduce a measure-valued Markov process modeling biochemical reactions in continuous space with both abundant and rare species. After suitable normalization, the system converges to a piecewise deterministic measure-valued process, and the

associated central limit theorem yields a semimartingale limit whose noise part can be interpreted as a conditionally piecewise Gaussian process.

How Differential Privacy Reshapes Optimal Statistical Design

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Much of the differential privacy literature focuses on inference under a fixed statistical design, but privacy constraints can drastically alter which study designs are optimal. I present results showing that designs that are statistically equivalent in classical settings may behave very differently under differential privacy, especially for functional data, with implications for how sampling resources should be allocated.

Distribution-Free and Model-Agnostic Changepoint Detection with Finite-Sample Guarantees

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We introduce ART, a distribution-free and model-agnostic framework for changepoint analysis with finite-sample guarantees. By transforming independent observations into exchangeable scores and aggregating ranked statistics, ART supports single and multiple changepoint testing, localization, and post-detection inference with exact finite-sample Type I error control.

High-dimensional Change-point Detection Using Generalized Homogeneity Metrics

Runmin Wang

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We study detection and localization of abrupt changes in high-dimensional distributions beyond the first two moments. The proposed nonparametric approach combines cumulative sum statistics in an embedded Hilbert space with wild binary segmentation, and is shown to perform well theoretically and empirically in high-dimensional settings.

On polynomial-time computation in PDE models

Sven Wang

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In complex statistical models involving PDEs, likelihood-based methods may face multimodal landscapes and computational intractability. We discuss a class of computationally tractable estimators for nonlinear inverse problems governed by elliptic PDEs and show that they achieve strong statistical guarantees while remaining polynomial-time computable.

A New Regression Lens on Multi-Class Classification

Marten Wegkamp

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We deepen linear discriminant analysis' connection to multivariate response regression by deriving an explicit relationship between discriminant directions and regression coefficients. This insight leads to a new regression-based procedure for multi-class classification that can flexibly incorporate structured, regularized, or nonparametric regression methods and is well-suited for establishing provable performance guarantees.

Can large language models boost the power of randomized experiments without statistical bias?

Waverly Wei

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We introduce CALM, a statistical framework that integrates large language model predictions with established causal estimators to increase precision in randomized experiments while preserving statistical validity. CALM treats large language model outputs as auxiliary prognostic information and corrects their potential bias through a calibration step, yielding efficiency gains over standard augmented inverse probability weighting estimators.

Statistical Inference under Adaptive Sampling with LinUCB

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Adaptively collected data has become ubiquitous in modern practice, yet even benign adaptive sampling schemes can introduce severe biases. Focusing on the linear bandit problem, we characterize the performance of LinUCB and derive inferential procedures that remain valid despite the challenges posed by adaptive data collection.

How do simple rotations affect the implicit bias of Adam?

Rebecca Willett

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Adaptive gradient methods such as Adam are sensitive to orthogonal transformations of feature space. We show that even small rotations of the data distribution can cause Adam to lose its richness bias and converge to inferior linear decision boundaries, and we discuss how an orthogonal reparameterization can restore rotational equivariance and richer solutions.

Gaussian Approximation and Concentration of Constant Learning-Rate Stochastic Gradient Descent

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We establish a comprehensive finite-sample and asymptotic theory for stochastic gradient descent with constant learning rates. Using a novel linear approximation technique, we derive quenched central limit theorems, Berry–Esseen bounds, and Nagaev-type tail inequalities for stochastic gradient descent iterates, providing rigorous foundations for uncertainty quantification.

Provably Reliable Classifier Guidance via Cross-Entropy Control

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We study classifier-guided diffusion models and show that controlling the cross-entropy loss of a probabilistic classifier at each diffusion step is sufficient to control the resulting guidance error. This yields quantitative bounds on sampling error and provides the first theoretical link between classifier training quality and effective guidance in diffusion models.

A modified Greenwood statistic for goodness-of-fit testing in heavy-tailed multivariate samples

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We introduce an enhanced version of the Greenwood statistic tailored to heavy-tailed multivariate data, including Pareto, Student's t , and alpha-stable distributions. The modified statistic improves goodness-of-fit testing and is able to determine whether a

sample originates from a distribution with infinite variance, as confirmed by simulations and real-data analyses.

Is the F-test doubly robust?

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We study the robustness of the F-test in random design linear models and show that its size remains close to nominal if either the normalized error distribution is close to spherical or the orthogonalized design matrix is close to Haar. At the same time, we identify limitations showing that this robustness can deteriorate rapidly as the setting departs from exact validity.

Model-Free Inference for High-Dimensional Binary Classification Using Repro Samples

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We develop a model-free inference framework for high-dimensional binary classification that leverages repro samples and a class of possibly misspecified sparse generalized linear models. The method provides guaranteed coverage for influential covariate sets and confidence sets for linear transformations of oracle regression coefficients, without requiring correct model specification or sparsity of the true model.

Identifiability and Inference for Generalized Latent Factor Models

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Generalized latent factor analysis provides a useful latent embedding approach across statistics, machine learning, psychometrics, econometrics, and social sciences. We establish inference properties for maximum likelihood estimators under commonly used identifiability conditions, including settings with correlated factors and non-orthogonal loading matrices, and illustrate the theory through simulations and a personality assessment application.

Detecting High-Dimensional Geometry in Random Graphs

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Random geometric graphs hide geometric structure inside an otherwise random-looking network. I will discuss recent results on detection thresholds for high-dimensional random geometric graphs, including smooth-kernel models and new progress on the sparse hard-threshold regime, together with a unifying conjecture linking the threshold to the standardized kernel operator.

Optimal Convex M-Estimation via Score Matching

Min Xu

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In the context of linear regression, we construct a data-driven convex loss function whose empirical risk minimization yields optimal asymptotic variance for estimating regression coefficients. The approach is motivated by score matching and log-concave projection, and leads to a computationally efficient semiparametric estimator that remains robust while sacrificing little statistical efficiency.

Causal Learning of Paired Vectors with Label Noise: Impact and Correction Methods

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Causal inference for paired vectors can be framed as a binary classification problem using kernel mean embeddings in feature space. We study the impact of mislabeled causal relationships on the training process and present robust learning methods that explicitly account for label noise, together with theoretical guarantees for their validity.

Outtrigger local polynomial regression

Elliot Young

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We introduce the outrigger local polynomial estimator, a distributionally adaptive modification of standard local polynomial regression that incorporates an estimate of the conditional score function of the errors. Subject to smoothness conditions, the method attains minimax-optimal performance up to a small multiplicative factor and can outperform the classical estimator outside the Gaussian setting.

On the hardness of group-conditional distribution-free predictive inference

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We study the limits of group-conditional coverage in distribution-free predictive inference, a relaxation of impossible pointwise conditional validity. We present hardness results characterizing when such guarantees can or cannot be achieved, and discuss implications for prediction with missing values, including an algorithm that attains informative predictive intervals conditional on missingness patterns.

Empirical Bayes for Dependent Data

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Empirical Bayes methods are traditionally developed under independence, but many modern applications involve dependent compound decision problems. We develop a nonparametric marginal likelihood approach under dependence, showing that the benefit of pooling is discounted by the largest eigenvalue of the correlation matrix and that nearly parametric oracle rates remain achievable for a variety of inferential targets.

Moments and Cumulants in Probability Theory: Recent Results and Perspectives

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Moments and cumulants are central objects in probability theory, especially in questions of determinacy and in limit theorems. I present recent results on cumulant-based limit theorems, moment-indeterminate distributions decomposable into Carleman-determinate parts, and new explicit bounds connecting moments and cumulants, together with broader perspectives on what can and cannot be recovered from one sequence to the other.

Convex Mixed-Integer Programming for Causal Additive Models with Optimization and Statistical Guarantees

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We study nonlinear additive structural equation models with Gaussian noise and formulate a maximum likelihood estimator with a group ℓ_0 penalty that directly controls graph sparsity. Despite the combinatorial nature of the problem, the estimator can be solved efficiently via a convex mixed-integer program combined with a novel outer approximation scheme, yielding globally optimal solutions together with consistency guarantees.

Statistical Inference in an Interactive Learning Paradigm

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We study generative models trained in interactive environments where training data arise not only from natural processes but also from other generative models. By formalizing these interactions through directed graphs, we derive explicit necessary and sufficient conditions for when model collapse occurs and characterize how network structure governs long-run behavior.

Priority queues with differing customer selection rules

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We consider priority queues in which each class may use its own customer selection rule, such as first-in-first-out or last-in-first-out. We show that the waiting time distribution for a given class depends only on the selection rule for that class and is insensitive to the rules used by other priority classes, leading to a general pathwise approach for deriving waiting time distributions.